MAHUM: A Multitasks Autoencoder Hyperspectral Unmixing Model

Jia Chen, *Member, IEEE*, Paolo Gamba, *Fellow, IEEE*, and Jun Li, *Fellow, IEEE*

***Abstract*— Hyperspectral unmixing is a crucial task in hyperspectral image processing and analysis. It aims to decompose mixed pixels into pure spectral signatures and their associated abundances. However, most current unmixing methods ignore the reality that the same pixel of a hyperspectral image has many different reflections simultaneously. To address this issue, we propose a multitask autoencoding model for multiple reflections, which can improve the algorithm's robustness in complex environments. Our proposed framework uses 3-D convolutional neural network (3DCNN)-based networks to jointly learn spectral-spatial priors and adapt to different pixels by complementing the advantages of other unmixing methods. The proposed method can quantitatively evaluate each area of data, which helps improve the algorithm's interpretability. This article presents a multitasks autoencoder hyperspectral unmixing model (MAHUM), which stacks multiple models to deal with various reflections of complex terrain. We also perform sensitivity analysis on some parameters and show experiment results demonstrating our method's ability to express the adaptability of different materials in different methods quantitatively.**

***Index Terms*— 3-D convolutional neural network (3DCNN), autoencoder (AE), hyperspectral imaging, multitask learning, remote sensing, spectral unmixing (SU).**

NOMENCLATURE

Symbol	Definition
$\mathbf{M}$	$(B \times R)$ endmember matrix with endmembers.
$\mathbf{B}$	Number of spectral bands.
$\mathbf{a}$	Corresponding abundance vector.
$\mathbf{n}$	$\mathbf{n}$ is an additive noise vector.
ψ_1, ψ_2, and ψ_3	Function constructed by our decoder's linear.
$\mathbf{V}_i^1\mathbf{h}_i^1$	Estimates of the endmembers weighted by the associated abundances.
$\mathbf{a}^i$	ith abundance.
$\mathbf{M}^i$	ith endmember.
φ_E	Encoder process.
φ_D	Decoder process.

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I. INTRODUCTION

HYPERSPECTRAL remote sensing data is characterized by nearly continuous spectral information, greatly enhancing its ability to detect and identify surface cover. As a result, it has become widely used in various applications such as land use analysis [1], [2], pollution monitoring [3], [4], wide-area reconnaissance [5], and field surveillance [6]. Due to the limitations of satellite technology and cost, hyperspectral data has high spectral resolution but low spatial resolution. This results in mixed pixels containing multiple materials that can be difficult to identify using spectral information alone. To address this issue, spectral unmixing (SU) has been developed to decompose mixed pixels into a set of pure spectral signatures and their associated abundances [7], [8].

SU typically begins with linear unmixing, which is based on the widely used linear mixture model (LMM) [7]. The LMM assumes that sunlight is reflected only once on the surface, and the observed spectrum is a linear combination of the endmembers. These algorithms aim to estimate the abundance fractions of each endmember in a mixed pixel and have various applications in remote sensing, such as land cover classification and mineral mapping. Algorithms based on the LMM model can be broadly categorized into five groups: geometry-based, nonnegative matrix factorization (NMF)-based, Bayesian-based, sparse-based, and autoencoder (AE)-based methods. Geometry-based methods: pixel purity index (PPI) [9], N-FINDR [10], and vertex component analysis (VCA) [11]. NMF-based methods: robust collaborative NMF [12], MVCNMF [13], $L_{1/2}$-NMF [14], sparsity-constrained deep NMF [15], and Self-Paced-NMF [16]. Bayesian-based methods: hierarchical Bayesian [17] and joint Bayesian [18]. Sparse-based methods: matching pursuit [19] and RGSU [20]. AE-based unmixing methods: UDAS [21] and DAEN [22]. However, the assumption of linear mixing needs to be revised in many complex scenarios. The reflection of light may occur twice or even more times in those scenarios.

Researchers have proposed the nonlinear mixture model (NLMM) to address more complex scenarios. NLMM is based on the hypothesis that materials undergo multiple scattering. Many scholars have considered intimate mixture materials as part of NLMM. An intimate mixture generally occurs at the micro-scale, which is much smaller than the path through which light passes. This phenomenon has been observed to impact hyperspectral image processing and

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analysis significantly. The solution for hyperspectral unmixing is typically based on Hapke, which includes methods such as multimixture pixel (MMP) [23] and linear mixture of intimate mixtures (LMIMs) [24]. For multilayered mixtures, corresponding models suggest that multiple interactions among scatterers occur at different layers, leading to an infinite sequence of powers of reflectance products. The generalized bilinear model (GBM) [25] is a popular method for unmixing hyperspectral images, which studies a GBM and a hierarchical Bayesian algorithm. Semi-NMF [26] presents a novel optimization method for nonlinear unmixing based on GBM, which is used to optimize the processing of an entire image in matrix form. In addition, robust NMF (rNMF) [27] introduces a powerful mixing model to describe hyperspectral data from the mixture of materials. Furthermore, several kernel-based nonlinear unmixing methods have been developed. Zhao et al. [28] proposed using archetypal analysis (KAA) for flexible endmember extraction, which implicitly considers intraclass variability in relative pure pixel mapping and mixed pixel unmixing. In the field of AE, deep multitask bilinear unmixing (DMBU) [29] develops an unsupervised unmixing approach based on the hierarchical bilinear mixture model (BMM). It solves two different tasks as a multitask and has shown promising results in hyperspectral image analysis. In addition, 3-D convolutional neural networks (3DCNNs) [30] use 3-D networks to increase spatial-spectral correlation and improve hyperspectral unmixing performance. The nonlinear unmixing method has successfully addressed the problem of multiple scattering to a large extent. However, current mainstream methods focus on solving the problem of parameter explosion caused by higher-order equations while ignoring the adaptability of a single material property to the algorithm. Zhao et al. [31] have considered the adaptability of various methods but have yet to evaluate them quantitatively. Quantitative evaluation of methods and materials can help reduce and update parameters and play an intuitive role in model analysis. Therefore, it is essential to develop quantitative evaluation methods for nonlinear unmixing algorithms to improve their performance and applicability in hyperspectral image analysis.

To address the adaptability of different materials to nonlinear unmixing algorithms, this article proposes a multitask autoencoder hyperspectral unsupervised unmixing model. The proposed model uses multiple unmixing models for various features to adapt to complex scenarios. In addition, the proposed method employs a multitask learning mode to update the weights between the methods. In experiment results, our method can quantitatively express the adaptability of different materials in different methods, demonstrating its effectiveness in hyperspectral image analysis.

The main contribution of this article is to use a multitasks autoencoder hyperspectral unmixing model to stack the model to deal with multiple reflections of complex terrain, the proposed method is named as MAHUM in this article. In summary, the contribution of this work can be concluded as follows.

1) A multitasks autoencoding model is proposed for multiple reflections, which can effectively improve the robustness of the algorithm in complex environments.
2) The proposed method can quantitatively evaluate each area of data, which helps to improve the interpretability of the algorithm.

These contributions demonstrate the effectiveness and applicability of our proposed method in hyperspectral image analysis.

The rest of this article is organized as follows. Some AE-based SU will be introduced in Section II. In Section III, the proposed method, namely, MAHUM, is presented in detail, and the main flowchart is given. The experiments and results are shown in Section IV. The conclusion is drawn in Section V.

II. Related Work

A. AE-Based SU

The work of this article is based on the AE network. The mixed pixels are decomposed using three different types of AE-based unmixing methods. Therefore, this section summarizes these three AE-based methods and analyzes their advantages.

AE-based unmixing methods have gained popularity in the SU field. AE is a neural network model for feature learning of input information. Its structure is divided into two parts: encoder and decoder. The function of the encoder is to encode the input data into a low-dimensional mapping space. At the same time, the decoder reconstructs the feature data in the mapping space to restore the original data. In this way, features obtained in the encoding process (which we call endmembers) and the weight in the decoding process (which we call abundances) can correspond to the unmixing process. Once launched, scholars applied it to the SU field and achieved good results. For example, [32] used a regular multiple hidden layer AE framework to conduct hyperspectral unmixing. Qu and Qi [21] adopted an l_{21}-norm on the encoder of the network to reduce the redundant endmembers while decreasing the reconstruction error (RE) simultaneously. Denoising-oriented unmixing [33] exploits the potential of AE to tackle the unmixing challenges. In [34], we take advantage of both automatic sampling and nonnegative sparse autoencoding and tackle problems with outliers. Specifically, the work in [34] imposed a one-norm to the latent output to generate sparse unmixing results. In [35], a new robust unmixing algorithm based on stacked nonnegative sparse AEs (NNSAEs) for hyperspectral data with outliers and low signal-to-noise ratio. In [22], we take advantage of the SAEs, it can unmix datasets with outliers and low signal-to-noise ratio. Convolutional neural network autoencoder unmixing (CNNAEU) [36] exploits the spatial and spectral structure of hyperspectral images both for endmember and abundance map estimation. Xu et al. [37] proposed a deep network for SU based on a new globallocal smoothing autoencoder. Jin et al. [38] proposed a novel technique network for unsupervised unmixing based on the adversarial AE.

Some excellent work is also being done to expand to the nonlinear field. For instance, DMBU [29] develops an

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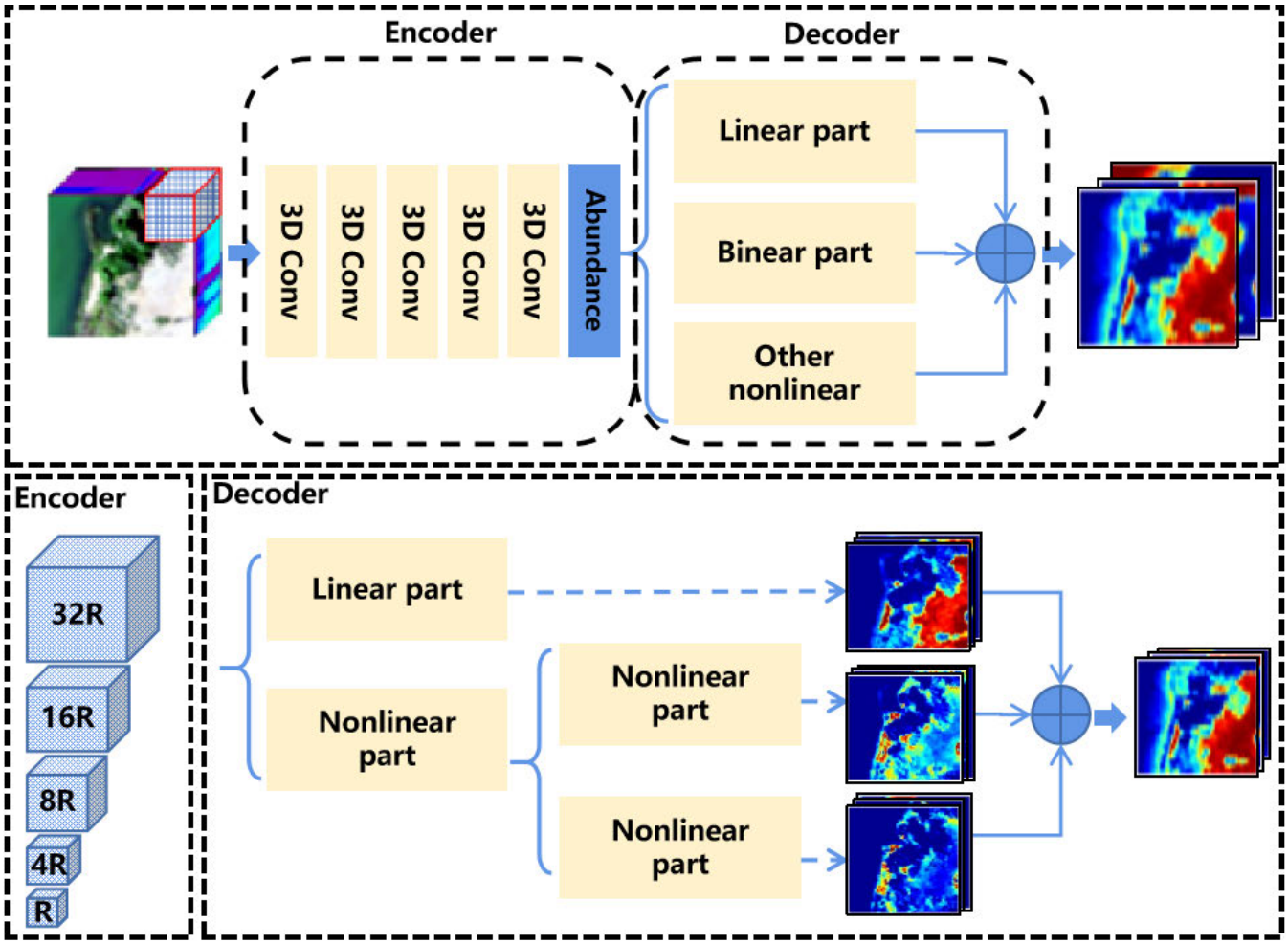


Fig. 1. Architecture of our proposed framework. It consists of two parts, an encoder and a decoder. In the encoding part, MAHUM uses a 3-D convolutional network, while in the decoding part, MAHUM uses a multitask framework. It should be noted that the method selection in each layer of the multitask framework depends on the initialized endmembers. This flowchart only takes water as an example. Therefore, the task combination of linear and nonlinear parts is carried out first. Then, there is the combination of bilinear and other nonlinear.

unsupervised unmixing approach based on the hierarchical BMM. Zhang et al. [39] designed a new deep autoencoder network structure based on the GBM to achieve nonlinear unmixing. It adds some regular constraint terms based on prior knowledge to avoid overfitting. In addition, 3DCNN [30] use 3-D networks to increase spatial-spectral correlation and improve hyperspectral unmixing performance. Long–short-term memory network and deep neural network (LSTM-DNN) [40] proposed a novel nonlinear unmixing scheme based on the LSTM-DNN to characterize the sequential property of a hyperspectral pixel. Wang et al. [41] proposed a novel scheme for blind nonlinear unmixing based on a carefully designed deep autoencoder network. Shahid and Schizas [42] developed a novel fully unsupervised autoencoder-based scheme for nonlinear SU by a unique averaging approach based on spatially aware filters built using radial basis function (RBF) kernels. Shahid and Schizas [43] combined a novel autoencoder structure with a kernelization layer, mapping the mixed pixels in a higher dimensional space for easier separability and a novel cross-product layer to account for nonlinear mixing mechanisms. Yang et al. [44] constructed a residual dense autoencoder network (RDAE) for nonlinear hyperspectral unmixing in multiple scattering scenarios. Hua et al. [45] presented an autoencoder network with adaptive abundance smoothing.

III. Multitasks Autoencoder Hyperspectral Unmixing Model

The research content of this stage is mainly aimed at the first research content. MAHUM is proposed. The main innovation of this model is to use more unmixing models for different features to adapt to complex features. In addition, the multitask model can help the model to be more robust. The main flowchart of the proposed MAHUM is shown in Fig. 1. In the encoder part, we use a five-layer 3-D convolution structure. Three-dimensional convolution can better mine spatial and spectral information in hyperspectral images. In the decoder part, we propose a multitask framework, hoping to improve the ability of pixel unmixing through the complementary form of multiple unmixing models. Besides, the symbol definitions used in this article are shown in Nomenclature.

A. Unmixing Framework

Our proposed approach employs three deep AEs to execute blind source separation on the observed data, bringing multiple hidden representations associated with the BMM.

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We first consider the LMM, where each observed pixel is assumed to be a linear combination of the endmembers weighted by their associated abundances

$$\mathbf{Y} = \mathbf{Ma} + \mathbf{n} \tag{1}$$

where $\mathbf{a}$ denotes the corresponding abundance vector, Boldface capital letters $\mathbf{Y}$ denote matrices, $\mathbf{M} = [\mathbf{m}_1, \ldots, \mathbf{m}_R]$ denotes the $(\mathbf{B} \times \mathbf{R})$ endmember matrix with endmembers $\mathbf{m}_i$, and $\mathbf{n} \in \mathbb{R}^B$ is an additive noise vector. As the abundances represent relative fractions of each material, they are required to satisfy the abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC), which are

$$\forall i : a_i > 0; \quad \sum_{i=1}^{R} a_i = 1. \tag{2}$$

The linear and bilinear components in the BMM correspond to the two hierarchies, respectively. It can be described as

$$Y = \mathbf{M}^1\mathbf{a}^1 + \mathbf{M}^2\mathbf{a}^2. \tag{3}$$

In this work, we consider the following general mixing mechanism:

$$Y = \psi_1\left(\mathbf{M}^1, \mathbf{a}^1\right) + \psi_2\left(\mathbf{M}^2, \mathbf{a}^2\right) + \psi_3\left(\mathbf{M}^3, \mathbf{a}^3\right) + \mathbf{n} \tag{4}$$

which consists of a linear mixture of endmembers $\mathbf{M}^1$ with abundance fractions $\mathbf{a}^1$, a bilinear fluctuation ψ_2 that defines the interactions of $\mathbf{M}^2$ parameterized by $\mathbf{a}^2$, and a nonlinear fluctuation ψ_3 that defines the interactions of $\mathbf{M}^3$ parameterized by $\mathbf{a}^3$. This form suggests that the nonlinear interactions of material signatures are in proportion to the abundance fractions of each material. This is reasonable. Because, for instance, a material with a negligible abundance will have limited contribution to either the linear component or the nonlinear component of $\mathbf{Y}$.

B. Encoder

Encoder φ_E compresses the input $\mathbf{Y}$ into a low-dimensional representation $\mathbf{h} \in \mathbb{R}^B$, that is,

$$\mathbf{h} = \varphi_E(\mathbf{Y}) \tag{5}$$

where $\varphi_E : \mathbb{R}^B \times 1 \Longrightarrow \mathbb{R}^B \times 1$. Recall that B represents the number of bands and R denotes the number of endmembers. Note that R is assumed priorly known in our work. There is a corresponding relationship between $\mathbf{h}$ and terminal element $\mathbf{a}$: $\mathbf{h} \Longrightarrow \mathbf{a}$.

To leverage spectral-spatial priors and improve unmixing performance, the hyperspectral cube is divided into N overlapping 3-D patches of size $n \times n \times B$, where n represents the spatial window size. The central pixel within each patch is considered the target pixel for decomposition. These patches are then inputted into our unmixing framework. The encoder and decoder can be either shallow or deep, but deeper networks are generally regarded as having better modeling capabilities.

This study employs two 3-D convolutional layers (3D Conv) (32R and 16R) with a spatial size of $3 \times 3 \times 8$ for capturing the spectral-spatial features. The spatial size of each patch is reduced to 1×1. Subsequently, three additional 3D Conv (8R, 4R, and R) are utilized to further capture the spectral-spatial features. The number of kernels gradually decreases in these layers, and the last layer has R kernels with an output feature dimension of $R \times 1$. There are no specific constraints imposed on the encoder to maximize network capacity usage and minimize information loss. Instead of using pooling layers, data compression is achieved through a stride strategy. All layers except the last hidden layer employ the same activation functions. The abundance maps are considered to satisfy two constraints, namely, ANC and ASC, which are consistent with the physical interpretation of the abundances.

C. Decoder

Decoder φ_D uncompresses the hidden representation vector $\mathbf{h}$ to reconstruct the original input data, that is,

$$\hat{\mathbf{Y}} = \varphi_D(\mathbf{h}) \tag{6}$$

where $\varphi_D : \mathbb{R}^B \times 1 \Longrightarrow \mathbb{R}^B \times 1$. The network trains the parameters and representations by minimizing the average RE between the input Y and its reconstructed counterpart $\hat{Y}$ given by

$$\mathcal{L}\left(\mathbf{Y}, \hat{\mathbf{Y}}\right) = \frac{1}{\mathbf{N}}\|\hat{\mathbf{Y}} - \mathbf{Y}\|^2 \tag{7}$$

where $\mathbf{N}$ is the number of pixels. With the output of the encoder $\mathbf{h} \in \mathbb{R}^B$, in this work the decoder is designed to reconstruct the input Y with the following specific structure:

$$\hat{\mathbf{Y}} = \psi_1\left(\mathbf{M}^1, \mathbf{a}^1\right) + \psi_2\left(\mathbf{M}^2, \mathbf{a}^2\right) + \psi_3\left(\mathbf{M}^3, \mathbf{a}^3\right) \tag{8}$$

where $\mathbf{V}$ is the weights of the first layer of the decoder. ψ_1, ψ_2, and ψ_3 are the function constructed by our decoder's linear, bilinear, and other nonlinear parts, respectively. There is a corresponding relationship between $\mathbf{V}$ and terminal element $\mathbf{M}$: $\mathbf{V} \Longrightarrow \mathbf{M}$.

To better deal with the three joint models of unmixing, we propose a multitask unmixing framework in this work. We adopted the multitask mode twice. The first group of complex terrain areas is divided into linear and nonlinear parts. The nonlinear part of the second group is divided into bilinear and other nonlinear parts. Equation (8) can be rewritten in this model as

$$\begin{cases} \mathbf{Y} = \psi_1\left(\mathbf{M}^1, \mathbf{a}^1\right) + \psi_{2,3}\left(\mathbf{M}^{2,3}, \mathbf{a}^{2,3}\right) + \mathbf{n}_1 \\ \psi^{2,3}\left(\mathbf{M}^{2,3}, \mathbf{a}^{2,3}\right) = \psi_2\left(\mathbf{M}^2, \mathbf{a}^2\right) + \psi_3\left(\mathbf{M}^3, \mathbf{a}^3\right) + \mathbf{n}_2. \end{cases} \tag{9}$$

This way, we can transform the three-mixed joint model into double mutually constrained two-mixed models. The multitask part can be shown in Algorithm 1 for a single two-mixed model.

When the initial value E_0A_0 is known, it is not difficult to calculate it through the iterative update rule of semi-NMF. This article recalled the operators and symbols defined in (4) and the decoder structure given by (8). The first layer of the decoder is then designed by

$$\mathbf{Y}_1 = \mathbf{V}^1\mathbf{h}^1 \tag{10}$$

where $\mathbf{V}^1$ is defined as the weights of the first layer of the decoder. If it is the first group of the linear and bilinear models,

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Algorithm 1 Multi-Task Model

Require: Image Y

Ensure: $\mathbf{M}^1, \mathbf{a}^1, \mathbf{M}^2, \mathbf{a}^2, \mathbf{M}^3, \mathbf{a}^3$ and $\mathbf{n}$

1: Some important parameters a.
   1) m: Number of endmembers.
   2) col, band: size of image.
   3) coeff, mu,eps, theta, lambda: some parameters of DMBU.
2: First multitask unmixing a.
   1) Run VCA and FCLS for the initialization. We can get the initialization value. $\mathbf{M}_0^1, \mathbf{a}_0^1$
   2) Use the initial value to obtain $\mathbf{M}_1^1, \mathbf{a}_1^1, \mathbf{M}_1^{2,3}, \mathbf{a}_1^{2,3}$ and $\mathbf{n}$ by DMBU's mutiltask mode. The specific solution process can be referred to [29]
   3) The output of the first part is $\mathbf{M}_1^1, \mathbf{a}_1^1$.
3: Second multitask unmixing a.
   1) The input is $Y - \mathbf{M}_1^1, \mathbf{a}_1^1$. The initialization value of the rest part is $\mathbf{M}_1^{2,3}, \mathbf{a}_1^{2,3}$.
   2) Use the initial value to obtain $\mathbf{M}_1^2, \mathbf{a}_1^2, \mathbf{M}_1^3, \mathbf{a}_1^3$ and $\mathbf{n}$ by DMBU's mutiltask mode.
   3) The output of the linear part is $\mathbf{M}_1^2, \mathbf{a}_1^2, \mathbf{M}_1^3, \mathbf{a}_1^3$ and $\mathbf{n}$.

endmember is extracted by linear algorithms. Like the VCA algorithm can be used to initialize the learning process of V^1

$$\mathbf{V}^1 = \text{blkdiag}\{\mathbf{V}_1^1, \ldots, \mathbf{V}_R^1\} \tag{11}$$

where blkdiag$*$ is a block diagonal matrix. Consequently, the product $\mathbf{V}^1\mathbf{h}^1$ equals to

$$\mathbf{V}^1\mathbf{h}^1 = \text{col}\{\mathbf{V}_1^1\mathbf{h}_1^1, \ldots, \mathbf{V}_R^1\mathbf{h}_R^1\}. \tag{12}$$

Vectors $\mathbf{V}_i^1\mathbf{h}_i^1$ are generated as estimates of the endmembers weighted by the associated abundances. The output $\mathbf{Y}_1$ of this layer is used as the input ψ_1 so that $\psi_1(\mathbf{V}^1\mathbf{h}^1)$ generates the linear component of the spectrum. $\mathbf{Y}_1$ is also used as the input of the bilinear component defined by a fully connected (FC) network without bias weights. This bilinear decoder component is designed to represent the nonlinear interactions among the endmembers weighted by the associated abundances.

In the linear part, we use the most common VCA model. In the bilinear part, we use the bilinear DMBU model. In the other nonlinear parts, we introduce a superficial hidden layer of deep learning. Among them, the hidden layer of the two layers can well use the feature of deep learning to analyze the remaining part. In addition, to ensure the validity of the results, we have used two different modes to determine the residual amount. Among them, the relative RE can make the results closer to the actual situation by ensuring the minimum of outliers. The minimum residual can obtain the minimum accuracy loss by providing the minimum residual.

D. Objective Function

Several components are considered to formulate the objective function of the proposed autoencoder. The mean-square error between the input and reconstructed data is employed for the data fitting

$$\mathcal{L}_{\text{data}}(\mathbf{Y}, \hat{\mathbf{Y}}) = \frac{1}{N}\sum_{i=1}^{N} \|\varphi_D(\varphi_E(\mathbf{Y}_i)) - \mathbf{Y}_i\|^2. \tag{13}$$

Blind unmixing problems with both endmember and abundance unknown can be a complex inverse problem. Regularization is often imposed to condition the problem with useful prior information. In this work, we first consider the regularity of the nonlinear function φ. Thus, the two-norm of the weights of the nonlinear part of the decoder (denoted by $\mathbf{V}_{\text{nlin}}$) given by

$$\mathcal{L}_{\text{reg}}(\mathbf{V}_{\text{nlin}}) = \| \mathbf{V}_{\text{nlin}}\|^2 \tag{14}$$

is used as the regularization to drive the weights to decay and avoid over-fitting. Furthermore, a first-order total variation norm (TV-norm) regularization is given by

$$\mathcal{L}_{\text{smo}}(\mathbf{V}^1) = \sum_{i=1}^{R}\sum_{j=1}^{B-1}\left|\left[\mathbf{V}_i^1\right]_{j+1} - \left[\mathbf{V}_i^1\right]_j\right|. \tag{15}$$

Such a regularization encourages the smoothness of the endmembers and reduces the estimation noise.

In our work, considering the influence of $\mathcal{L}_{\text{smo}}(\mathbf{V}^1)$ on parameters, we designed two different objective functions from the perspective of minimum residual and minimum outlier

$$\mathcal{L}_1 = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{reg}} \tag{16}$$

$$\mathcal{L}_2 = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{smo}}. \tag{17}$$

Considering the adaptability of algorithms to different application scenarios. This article proposes two different methods, MAHUM-1 and MAHUM-2, based on the aforementioned losses and (9). The optimization function of (9) is described as

$$\psi_{2,3}(\mathbf{M}^{2,3}, \mathbf{a}^{2,3}) = \arg\min_{\text{ANC,ASC}} \mathbf{Y} - \psi_1(\mathbf{M}^1, \mathbf{a}^1). \tag{18}$$

Here, if we take the minimum residual (MAHUM-1) for (18), the algorithm will have less information loss, but its adaptability to complex scenarios will become weaker. When using the minimum outlier (MAHUM-2), the algorithm's adaptability to complex scenarios becomes stronger. Please refer to DMBU [29] for details on this section.

IV. Experiments

To verify the accuracy of the proposed algorithm, this section will be described in the following sections: 1) datasets; 2) quantitative evaluating indicators; 3) comparison with state-of-the-art methods; 4) experiments with synthetic data; and 5) experiments with real data.

A. Datasets

Quantitative evaluation data results will be displayed in several datasets. This includes both synthetic and real datasets.

The synthetic dataset randomly selects three pure spectral features from the United States Geological Survey (USGS) spectral library. Each pixel in the dataset has 224 spectral

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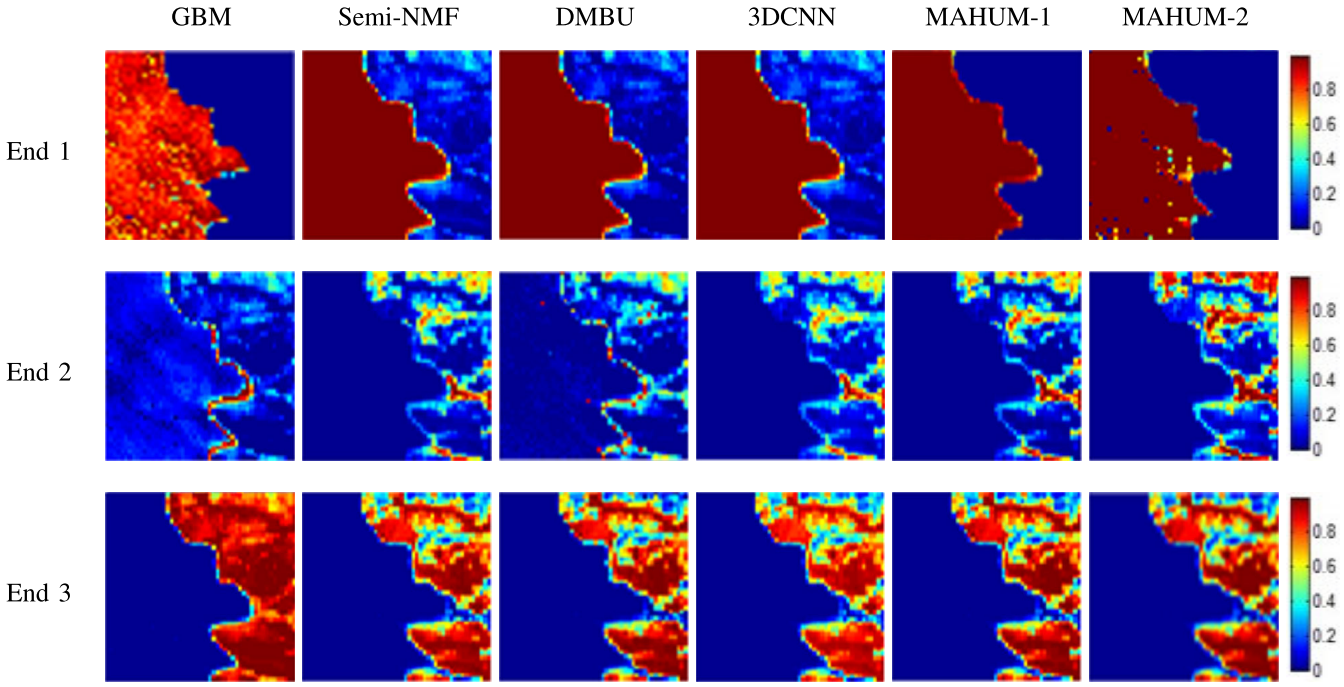


Fig. 2. Abundance maps of the water, soil, and tree from the Moffett Field dataset obtained by the different algorithms.

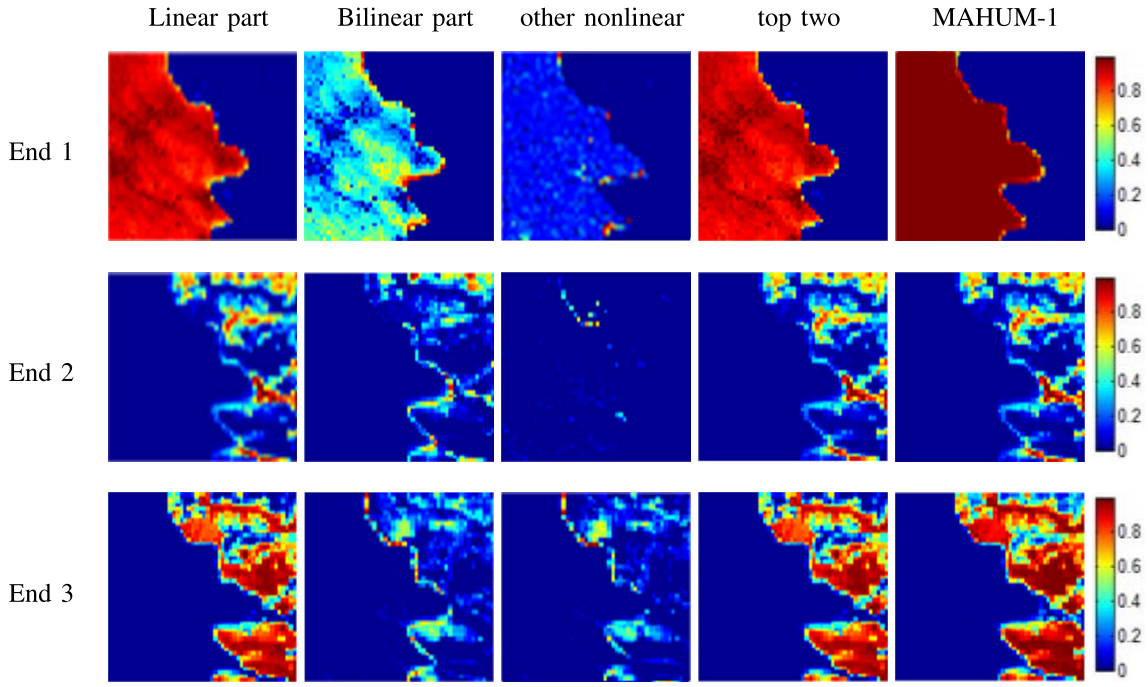


Fig. 3. Moffett Field dataset (MAHUM-1). The linear, bilinear, and other nonlinear parts are the quantitative analysis results of the three types of unmixing methods proposed in this article. (Top two) Result of adding the two items with the most unmixing components according to their weights. MAHUM refers to the result of all unmixing components.

bands, covering a spectral range of 0.4–2.5 m. Abundance settings follow a random distribution. The synthetic dataset is generated using a BMM. The abundance settings adhere to a random distribution, and the highest level of purity in the synthetic data's abundance is set at 0.9.

Mainstream datasets include Moffett Field, Jasper Ridge, and Samson. In addition, this article specifically tested hyperspectral datasets based on the University of Pavia in Italy for complex urban scenarios.

The Moffett Field data is an actual hyperspectral image acquired over Moffett Field, CA, USA, in 1997. It was gathered by the Airborne Visible Infra-Red Imaging Spectrometer (AVIRIS). The data size in the experiment is 50×50 pixels. It has 189 bands. The wavelength range is 0.4–2.5 m.

The Jasper Ridge data was gathered by AVIRIS over Jasper Ridge, CA, USA. The data size in the experiment is 100×100 pixels. It has 198 bands. The wavelength range

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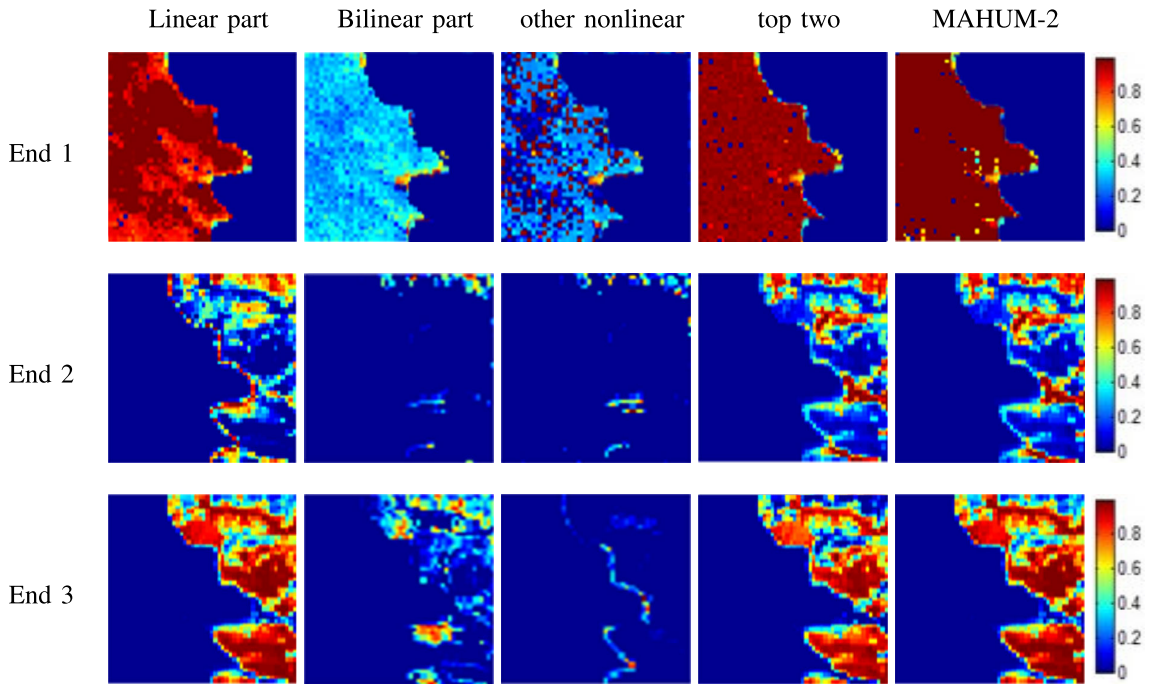


Fig. 4. Moffett Field dataset (MAHUM-2). The linear, bilinear, and other nonlinear parts are the quantitative analysis results of the three types of unmixing methods proposed in this article. (Top two) Result of adding the two items with the most unmixing components according to their weights. MAHUM refers to the result of all unmixing components.

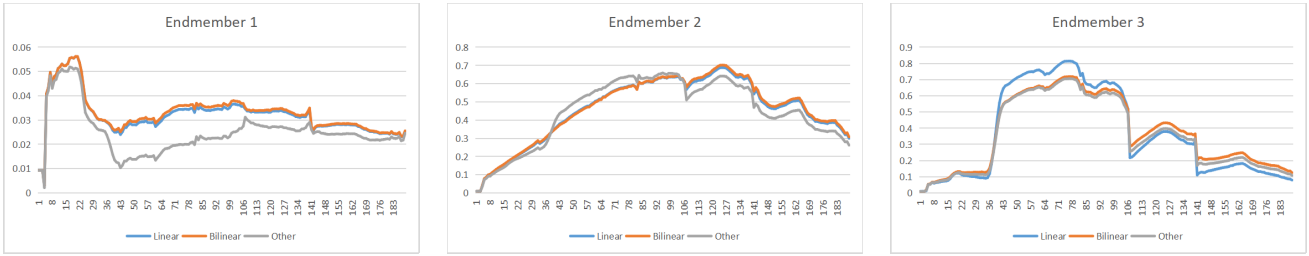


Fig. 5. Moffett Field dataset: estimated nominal endmember signatures.

is 0.38–2.5 m. This dataset has four abundances (water, soil, tree, and road).

The Samson dataset was obtained by the SAMSON sensor. The data size in the experiment is 95 × 95 pixels. It has 156 bands. The wavelength range is 0.401–0.889 m. the Pavia data is a small portion of the scene acquired by the PRISMA sensor operated by the Italian Space Agency (ASI) in April 2021. The dataset consists of 199 bands and covers the whole city of Pavia with only 120 × 120 pixels. Indeed, given the spatial resolution of PRISMA data (30 m), this urban dataset is challenging and characterized by the fact that the features of the urban scene are narrow and difficult to extract. This is true for the river, for instance. In addition, roads and buildings are intertwined, forming a clear discrete situation, which makes it difficult to accurately extract them.

B. Quantitative Evaluation Indicators

Four metrics are adopted for the assessment of results [46], including root mean square error (RMSE), the spectral angle distance (SAD), the mean SAD (mSAD), and the reconstructed error (RE).

The first evaluation indicator is RMSE, which is used to evaluate abundance performance. The value of RMSE represents the deviation of the predicted value from the true value. The smaller the RMSE, the closer the unmixing result is to the true value.

The second evaluation indicator is SAD. SAD evaluates the difference between each endmember and the actual value. The smaller the SAD value, the smaller the spectral difference between the obtained endmember and the true value.

The third evaluation indicator is mSAD, which is the mean value of SAD. It reflects the difference between the overall endmember and the true value. Its numerical performance is consistent with SAD.

The fourth evaluation indicator is RE, which measures data fitting. The smaller the value of RE, the better the fit between the results and the reality.

C. Comparison With State-of-the-Art Methods

MAHUM is evaluated by comparing it with several widely used unmixing algorithms. These algorithms include N-FINDR [10], VCA [11], FCLS [47], $L_{1/2}$-NMF [14], uDAS [21], GBM [25], Semi-NMF [26], rNMF [27], DMBU [29], and 3DCNN [30]. among them, N-FINDR and VCA are just used for extracting endmember. FCLS, GBM, and Semi-NMF can only be used for obtaining abundance fractions. Therefore,

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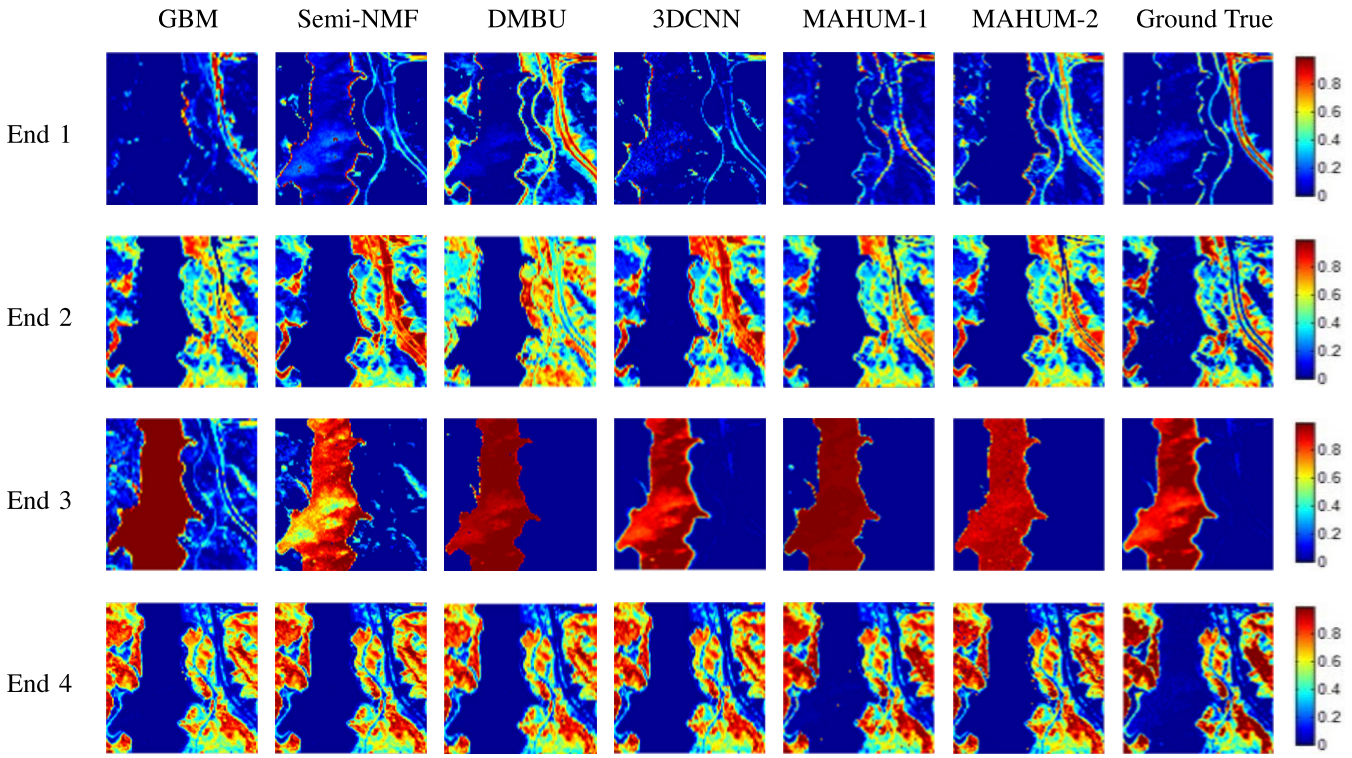


Fig. 6. Abundance maps of the water, soil, and tree from the Jasper Field dataset obtained by the different algorithms.

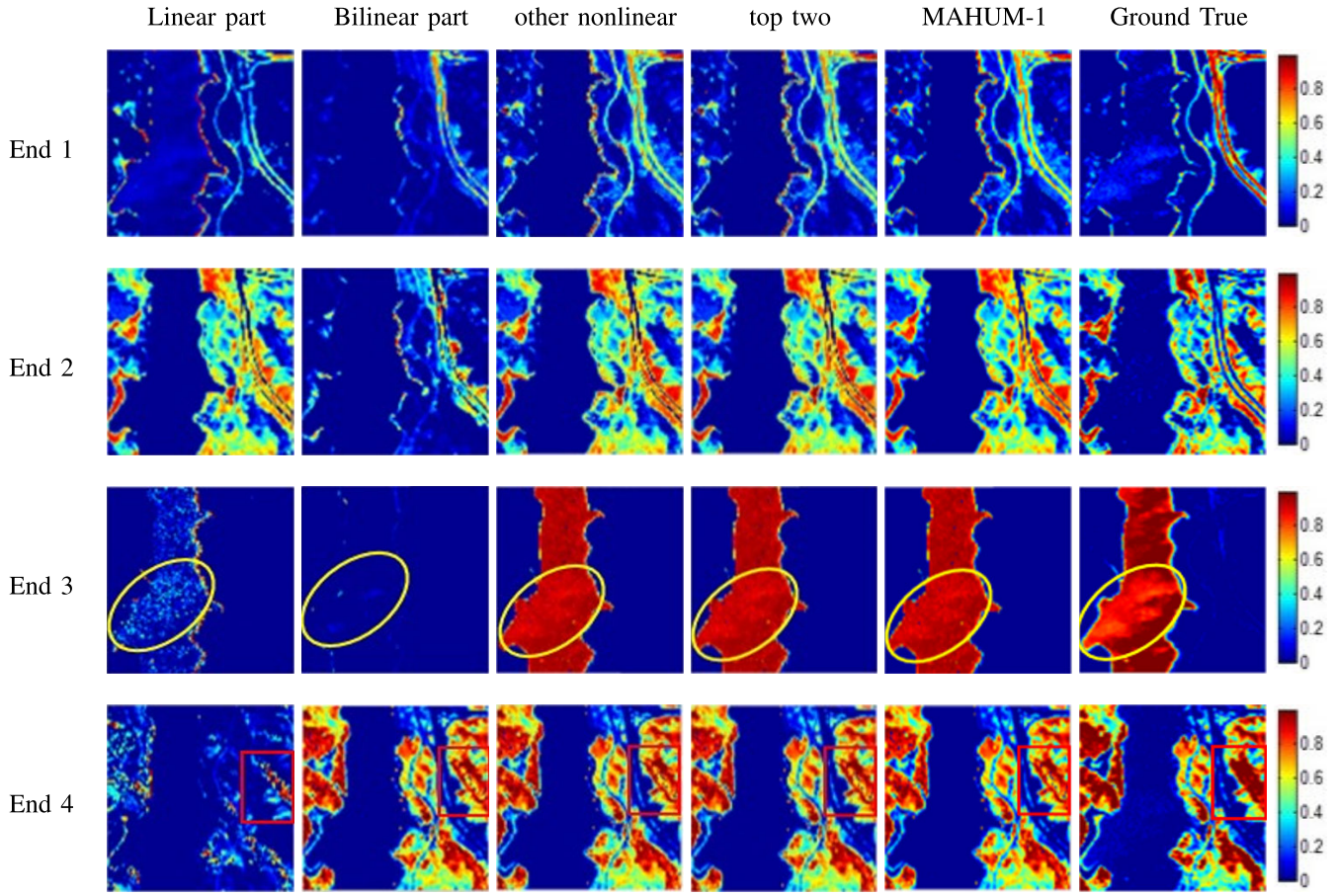


Fig. 7. Jasper Field dataset (MAHUM-1). The linear, bilinear, and other nonlinear parts are the quantitative analysis results of the three types of unmixing methods proposed in this article. (Top two) Result of adding the two items with the most unmixing components according to their weights. MAHUM refers to the result of all unmixing components.

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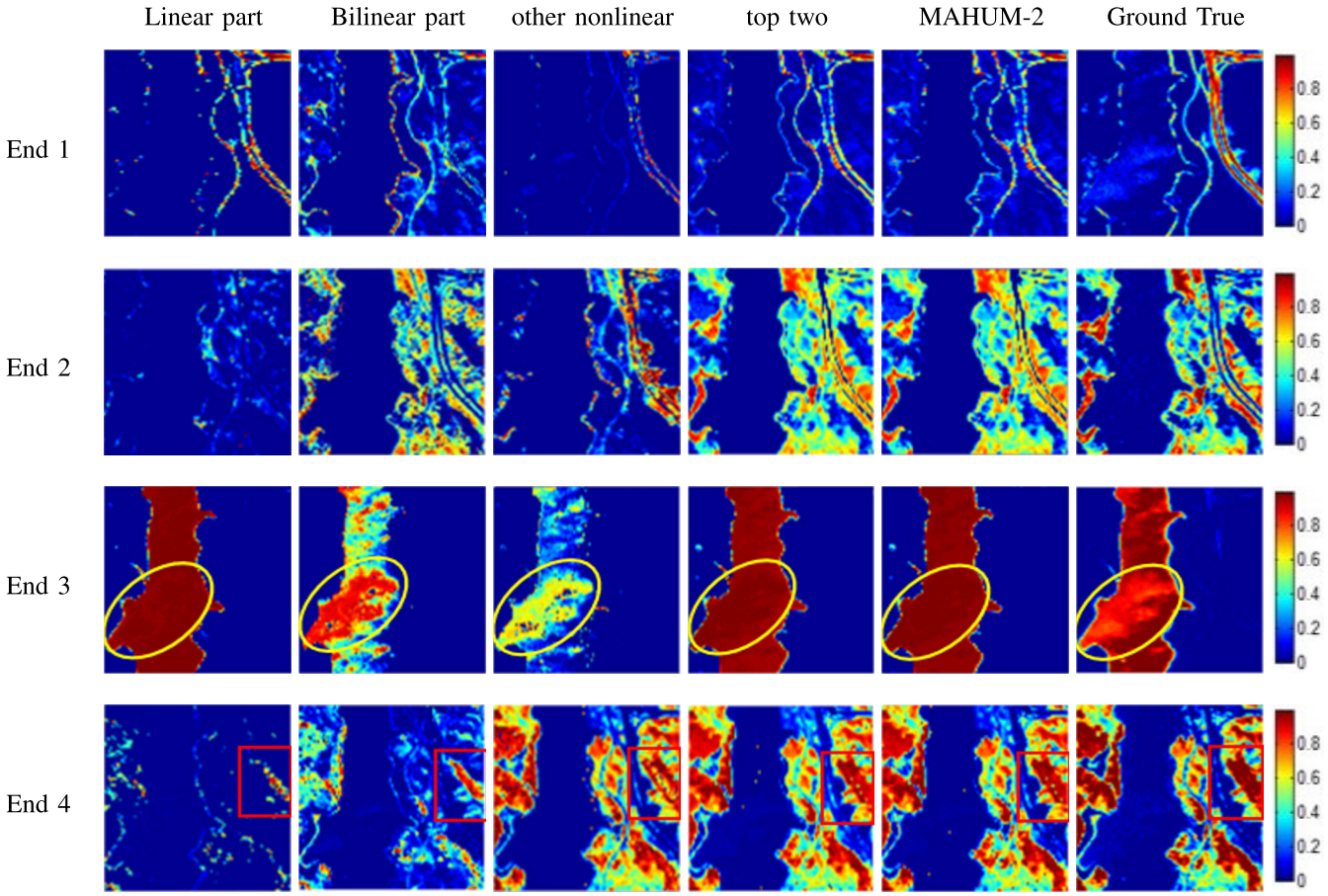


Fig. 8. Jasper Field dataset (MAHUM-2). The linear, bilinear, and other nonlinear parts are the quantitative analysis results of the three types of unmixing methods proposed in this article. (Top two) Result of adding the two items with the most unmixing components according to their weights. MAHUM refers to the result of all unmixing components.

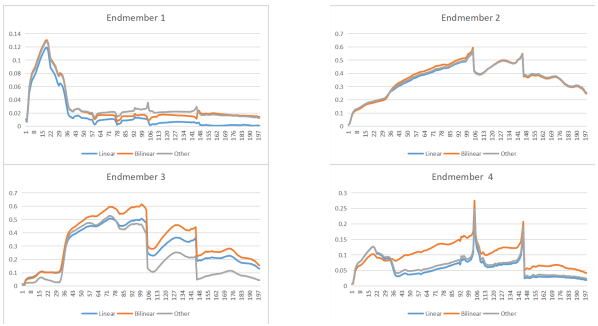


Fig. 9. Jasper Field dataset: estimated nominal endmember signatures.

they need to be combined in the experiment. It should be noted that VCA, N-FINDR, FCLS, uDAS, and $L_1/2$-NMF are approaches for linear unmixing. On the contrary, GBM, Semi-NMF, rNMF, DMBU, and 3DCNN are nonlinear unmixing methods.

D. Experiments With Synthetic Data

The quantitative results are shown in Table I. The results reported in Table I show that the proposed MAHUM obtains competitive results when compared with other methods. More specifically, the proposed approach can get good results in estimating endmembers and abundances. This is demonstrated by the fact that it achieves better results in terms of SAD, RMSE, and RE. Some important parameters were set in this experiment: DMBU: $\mu = 0.5$ and $v = 0.1$. 3DCNN: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^4$, and $\gamma = 1 \times 10^6$. MAHUM: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^4$, $\mu = 0.5$, $v = 0.1$, and $\gamma = 1 \times 10^6$.

The results reported in Table I show that the proposed MAHUM obtains competitive results when compared with other methods. More specifically, the proposed approach can get good results in estimating endmembers and abundances. This is demonstrated by the fact that it achieves better results regarding RMSE and RE. Our method is also, on average, in the evaluation of SAD.

E. Experiments With Real Data

In this section, the proposed MAHUM is applied to three real hyperspectral datasets: Moffett Field, Jasper Ridge, Samson, and Pavia.

1) Experiment With Moffett Field Dataset: Fig. 2 shows the comparison between our method and other methods with Moffett data. From the visual effect shown in Fig. 2, most methods have defects in water performance. Among them, GBM performed the worst. Overall, all methods have shown their advantages for soil. The method proposed in this article significantly improves all three endmembers, especially for the performance of trees. Some important parameters were set in this experiment: DMBU: $\mu = 0.5$ and $v = 0.2$. 3DCNN: the

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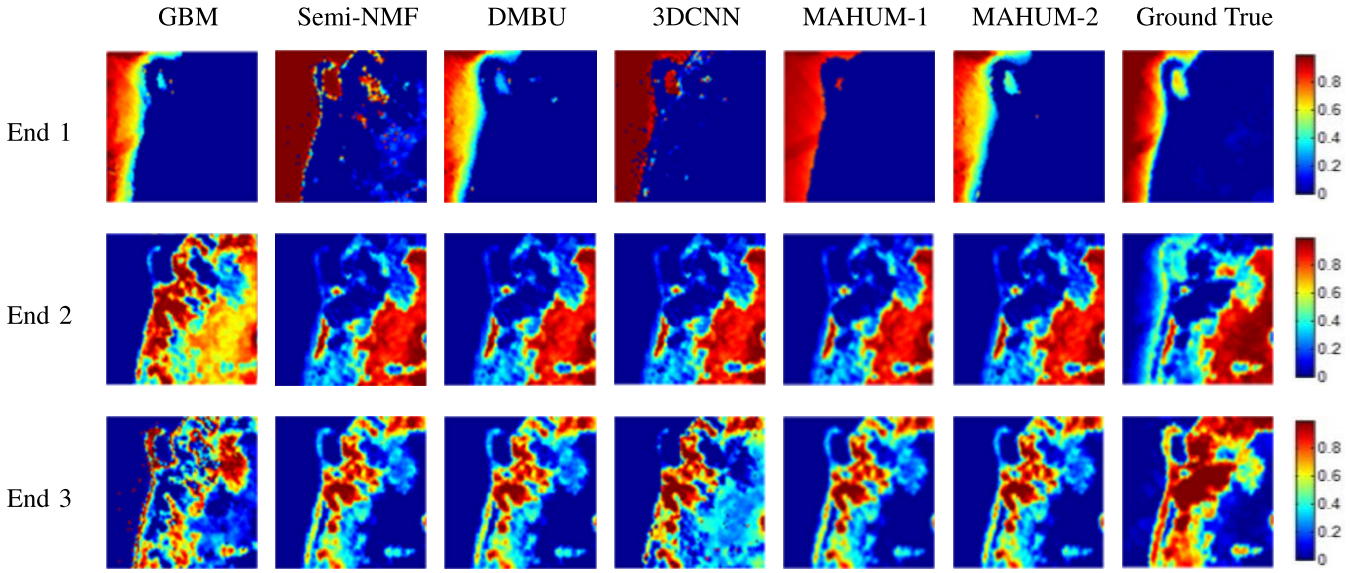


Fig. 10. Abundance maps of the water, soil, and tree from the Samson Field dataset obtained by the different algorithms.

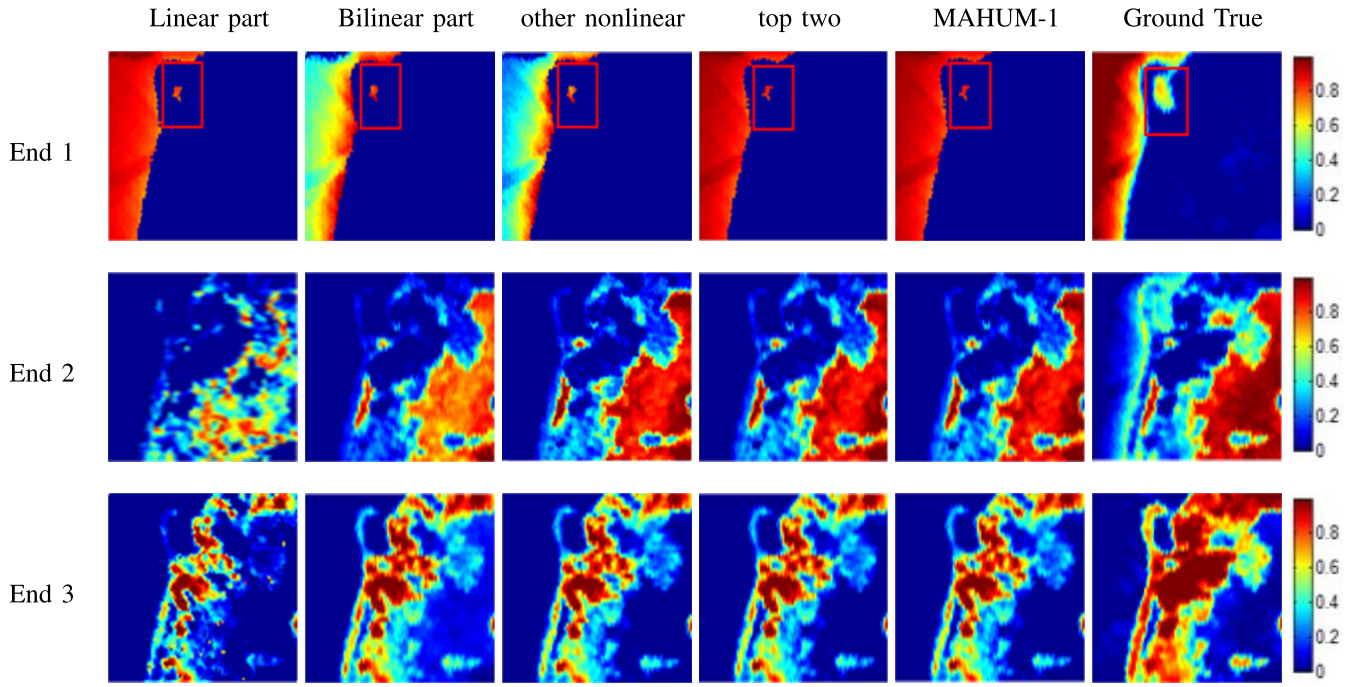


Fig. 11. Samson dataset (MAHUM-1). The linear, bilinear, and other nonlinear parts are the quantitative analysis results of the three types of unmixing methods proposed in this article. (Top two) Result of adding the two items with the most unmixing components according to their weights. MAHUM refers to the result of all unmixing components.

TABLE I
SYNTHETIC DATA, SNR = 20 dB

	N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU	3D-CNN	MAHUM-1	MAHUM-2
mSAD	0.0139	0.0096	0.0305	0.0104	0.0094	0.0106	0.0093	0.0098	0.0106	0.0102	**0.0092**
RMSE	0.3896	0.3722	0.3903	0.3709	0.3798	0.3719	0.3677	0.3652	0.3762	0.3641	**0.3637**
RE	0.0137	0.0152	0.0225	0.0138	0.0145	0.0153	0.0147	0.0127	0.0136	0.0118	**0.0112**

learning rate was set to 1×10^3, $\lambda = 1 \times 10^6$, and $\gamma = 1 \times 10^6$. MAHUM: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^6$, $\mu = 0.5$, $v = 0.2$, and $\gamma = 1 \times 10^6$.

Table II shows that the proposed MAHUM-1 obtained the best RE value. From a numerical perspective, the two methods presented in this article are clearly in the leading position. However, MAHUM-2 exhibits noise in the water. An extensive residual setting causes this.

Figs. 3 and 4 show the advantages of MAHUM-1 and MAHUM-2, respectively. As can be seen from Figs. 3 and 4, our algorithm uses three different algorithms to evaluate the characteristics of unmixing quantitatively. And they all

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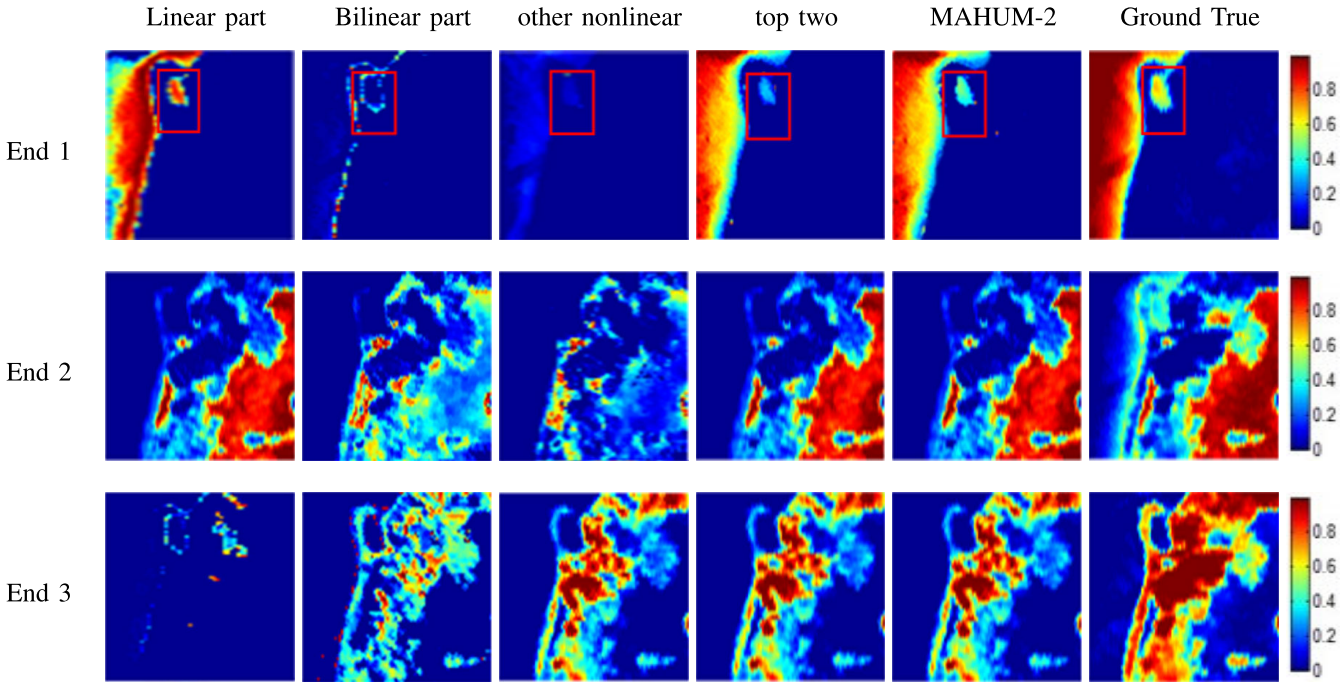


Fig. 12. Samson dataset (MAHUM-2). The linear, bilinear, and other nonlinear parts are the quantitative analysis results of the three types of unmixing methods proposed in this article. (Top two) Result of adding the two items with the most unmixing components according to their weights. MAHUM refers to the result of all unmixing components.

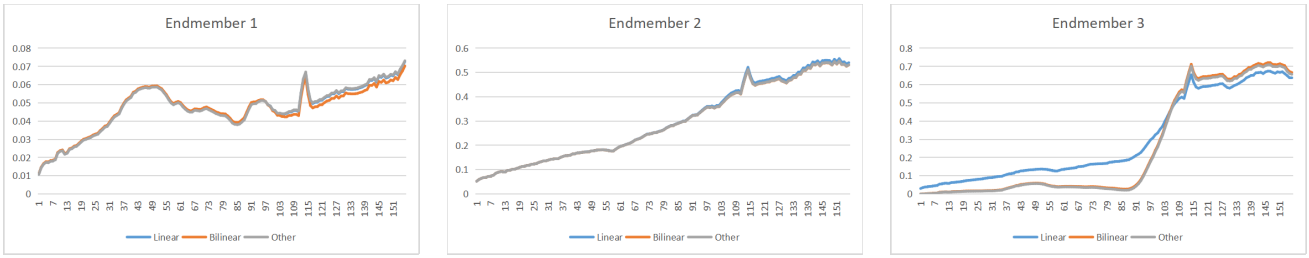


Fig. 13. Samson Field dataset: estimated nominal endmember signatures.

TABLE II
MOFFETT DATASET

	N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU	3D-CNN	MAHUM-1	MAHUM-2
RE	0.0265	0.0258	0.0255	0.0239	0.0264	0.0295	0.0194	0.0196	0.0213	**0.0173**	0.0195

achieved their expected goals. The results show that, for example, water has more linear components, but there is still a certain degree of nonlinear unmixing. This is consistent with the actual situation. In addition, as shown in the red boxes in Figs. 3 and 4. For environments such as soils, where there have been many nonlinear changes, some parts still undergo linear unmixing, while the nonlinear parts exhibit little. This is consistent with the actual situation. It also shows that our method can adaptively exploit the advantages of various methods for different endmember components.

From Fig. 5, it can be seen that the three end components of Moffett data have high fidelity in this algorithm, and the end component spectra in the three methods remain basically consistent.

2) Experiment With Jasper Ridge Dataset: Some important parameters were set in this experiment: DMBU: $\mu = 0.5$ and $v = 0.1$. 3DCNN: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^4$, and $\gamma = 1 \times 10^6$. MAHUM: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^4$, $\mu = 0.5$, $v = 0.1$, and $\gamma = 1 \times 10^6$.

Table III shows that the proposed DMBU obtained the best RMSE and RE scores in this experiment. Moreover, the quantitative results illustrate that the proposed MAHUM obtained better SADs. For illustrative purposes, the endmember signatures and abundance maps are shown in Fig. 6, respectively. Figs. 7 and 8 show the abundances estimated via MAHUM-1 and MAHUM-2, and it can be seen that these abundances match well with the corresponding references. Figs. 7 and 8 reveal that the endmembers estimated via MAHUM also match the corresponding library signatures. The results in Table III and Fig. 7 further demonstrate the effectiveness of the linear part of the proposed MAHUM-1. On the other hand, Fig. 8 shows that MAHUM-2 also performs well in the minimum residual.

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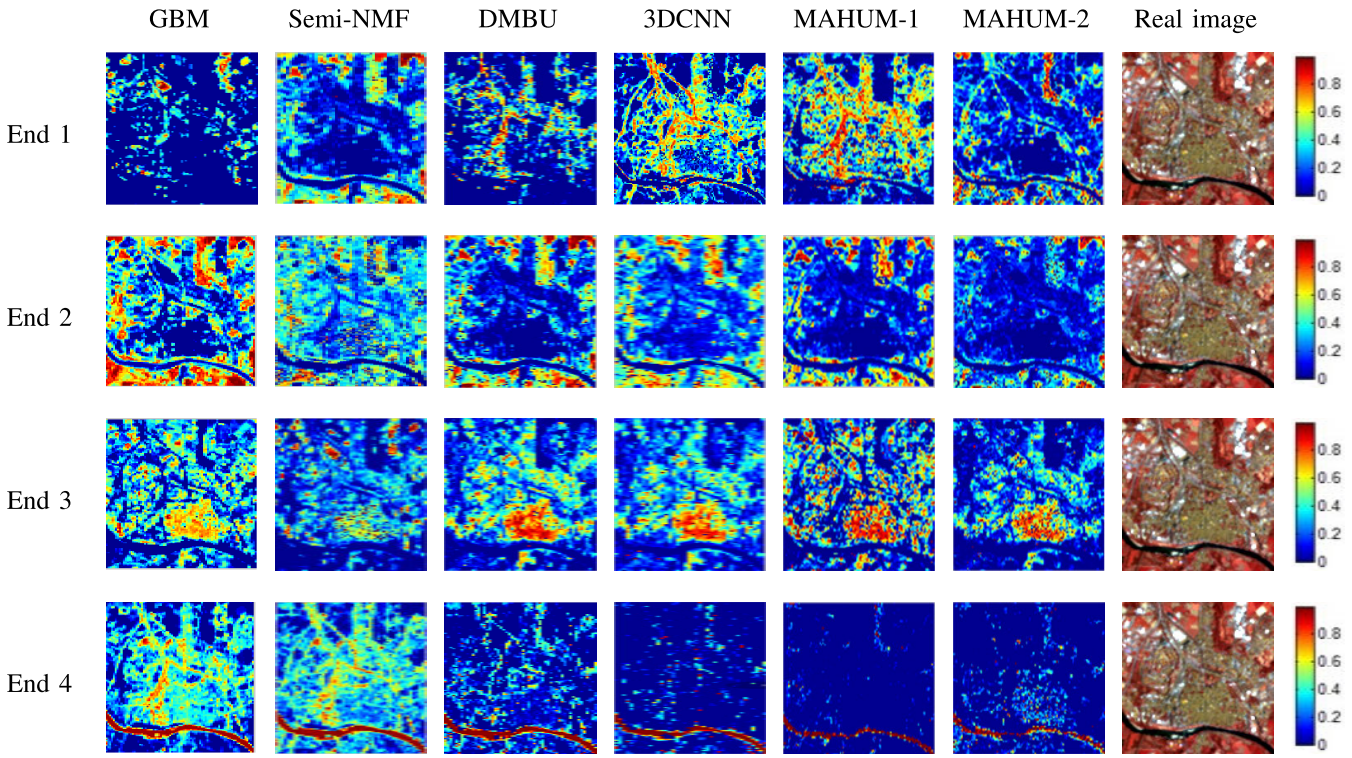


Fig. 14. Abundance maps of water, soil, buildings, and trees extracted from the Pavia PRISMA dataset by different algorithms.

TABLE III
JASPER DATASET

	N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU	3D-CNN	MAHUM-1	MAHUM-2
SAD (Road)	0.0964	0.0942	0.0948	0.0742	0.0793	0.0859	0.0849	0.0653	0.0749	**0.0529**	0.0549
SAD (Soil)	0.1496	0.1207	0.1506	0.1184	0.1053	0.0959	0.0765	0.0698	0.0745	**0.0604**	0.0649
SAD (Water)	0.1309	0.1549	0.1136	0.1074	0.1086	0.0958	0.0854	0.0804	0.0976	**0.0715**	0.0934
SAD (Tree)	0.1953	0.1833	0.1504	0.1693	0.1407	0.1594	0.1697	0.1128	0.1269	0.0931	**0.0879**
mSAD	0.1430	0.1382	0.1273	0.1173	0.1084	0.1092	0.1041	0.0820	0.0934	**0.0571**	0.0593
RMSE	0.1723	0.1772	0.1927	0.1820	0.1501	0.1620	0.1724	0.1541	0.1645	**0.1368**	0.1463

Our experiment results obtained the best results on the Jasper dataset. In addition, we can find that the experiment results fully meet the expectations from Figs. 7 and 8. That is, we can decompose the same region through various unmixing methods. In this way, the characteristics of each method can be brought into play to obtain better results. In addition, Figs. 7 and 8 can also reflect the advantages of this algorithm in adaptive and quantitative analysis. First of all, MAHUM can select appropriate unmixing methods for endmembers. For example, the red box in Figs. 7 and 8. Soil is generally dominated by multiple scattering, so its unmixing type should be dominated by nonlinearity. Our algorithm also indicates this situation. Most of the unmixing results are concentrated in the nonlinear part. In addition, in the red box section, a phenomenon needs attention, although most of the information is detected and successfully unmixed in other nonlinear sections. However, there are still a small number of occurrences in the linear and bilinear parts. Fig. 7 mainly focuses on the bilinear part, while Fig. 8 includes both linear and bilinear parts. This indicates that this part of the soil is adaptable to both linear and bilinear unmixing methods. In the yellow box section, we can see from the ground truth that although it is all water, there is still a different section from other water areas in the yellow box section. From our experiment results, our algorithm is very obvious for detecting this region block. Although most of the linear parts have been seen, it still exists in bilinear and other nonlinear aspects.

For the endmembers, it can be seen from Fig. 9 that the spectral errors of the four endmembers of Jasper data in all three methods are within a controllable range. Retained the complete spectral characteristics of the endmembers.

3) Experiment With Samson Ridge Dataset: Some important parameters were set in this experiment: DMBU: $\mu = 0.5$ and $v = 0.1$. 3DCNN: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^6$, and $\gamma = 1 \times 10^6$. MAHUM: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^6$, $\mu = 0.5$, $v = 0.1$, and $\gamma = 1 \times 10^6$.

Table IV shows the comparison results between the Samson Ridge dataset and other classical methods. There are a total of four endmembers with complex urban environments and various land features intertwined in this dataset. This has resulted in many methods not being effective on this dataset. However, from a numerical perspective, our proposed method is basically ahead of other methods in terms of accuracy. Many methods lack robustness for such complex scenarios.

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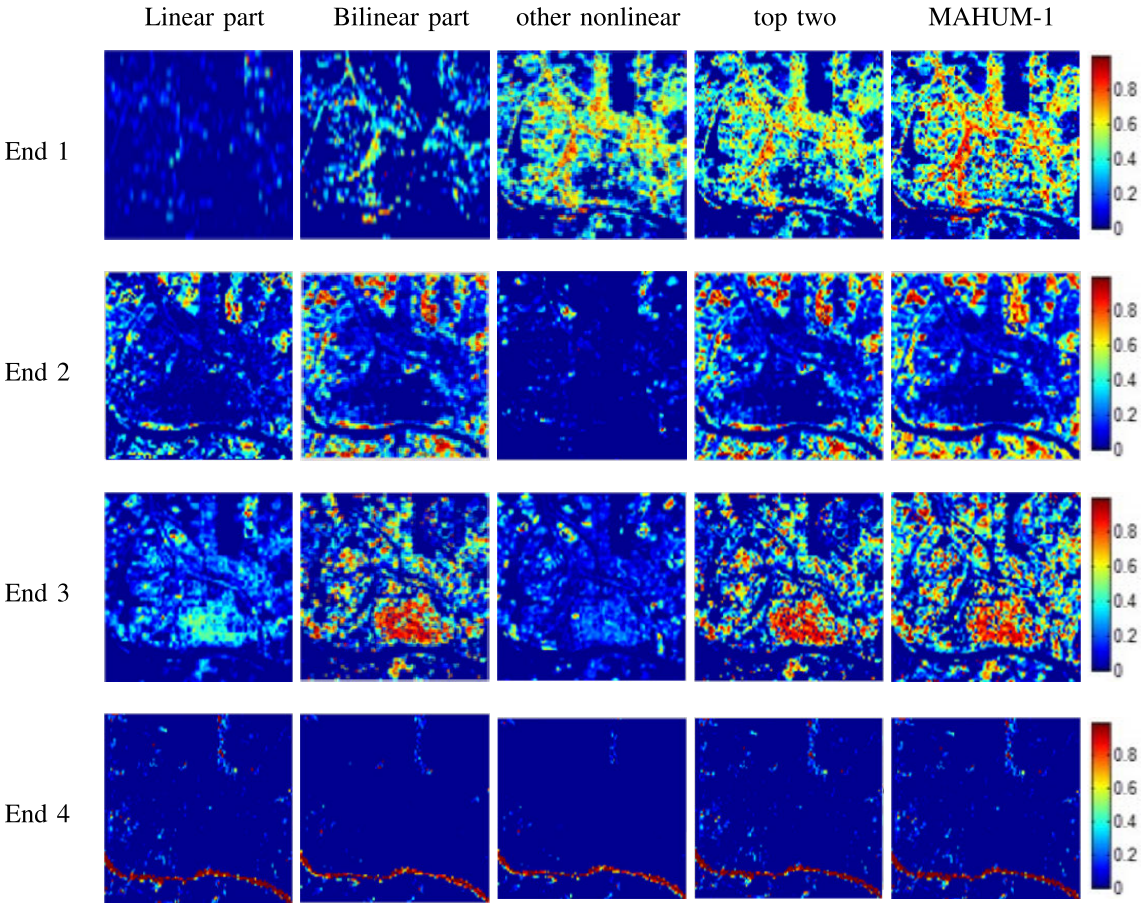


Fig. 15. Results of the analysis of the Pavia PRISMA dataset by means of the MAHUM-1 algorithm. The linear, bilinear, and other nonlinear parts are the quantitative analysis results of the three types of unmixing methods proposed in this article. (Top two) Result of adding the two items with the most unmixing components according to their weights. MAHUM refers to the result of all unmixing components.

TABLE IV
SAMSON DATASET

	N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU	3D-CNN	MAHUM-1	MAHUM-2
SAD (Water)	0.0582	0.0457	0.0385	0.0287	0.0472	0.0380	0.0361	0.0227	0.0239	0.0279	**0.0185**
SAD (Soil)	0.1735	0.1325	0.1693	0.1832	0.1174	0.1377	0.1480	0.0975	0.0964	0.0915	**0.0844**
SAD (Tree)	0.0578	0.0489	0.0573	0.0498	0.0753	0.0433	0.0378	0.0219	0.0291	0.0230	**0.0172**
mSAD	0.0838	0.0844	0.0941	0.0854	0.0690	0.0890	0.0954	0.0638	0.0521	0.0483	**0.0396**
RMSE	0.3729	0.3863	0.3682	0.3651	0.3312	0.3074	0.3029	0.2795	0.2597	0.2368	**0.2073**

For example, GBM and semi-NMF. In addition, 3DCNN also performs well in this scenario.

Fig. 10 shows several classical methods' comparison and display results. From the visual effect, the performance of water is not very good. However, the results of DMBU and MAHUM-2 are closer to the ground truth. Semi-NMF performed the worst. On the soil, GBM has encountered many misclassifications. Its effect is the worst. The results of the other methods are similar visually. In terms of tree performance, 3DCNN did not meet expectations. Several other methods behave similarly.

Figs. 11 and 12 are diagrams showing the quantitative results of MAHUM-1 and MAHUM-2. From Figs. 11 and 12, it can be seen that the allocation of the MAHUM-1 methods and the three endmembers in the three methods are similar to the actual situation. For example, water exhibits more linear unmixing, while trees and soil show more nonlinear changes. In the red box, we can see that MAHUM-1 can be detected in all three method sections. MAHUM-2 only appears in partial linear decomposition, indicating that our algorithm can optimize the algorithm process by modifying constraint terms. MAHUM-2 is closer to the actual situation.

Fig. 13 shows the endmembers extracted by different methods. We can see that the endmembers of the three methods remain basically unchanged.

4) Experiment With Pavia PRISMA Dataset: Some important parameters were set in this experiment: DMBU: $\mu = 0.5$ and $v = 0.3$. 3DCNN: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^4$, and $\gamma = 1 \times 10^3$. MAHUM: the learning rate was set to 1×10^3, $\lambda = 1 \times 10^4$, $\mu = 0.3$, $v = 0.1$, and $\gamma = 1 \times 10^3$.

Table V shows the comparison results between the Pavia PRISMA dataset and other classical methods. There are three endmembers in the Samson dataset, including soil, trees, and water. From the numerical results, MAHUM-2 performs best. DMBU and 3DCNN performed well on some effects, such as

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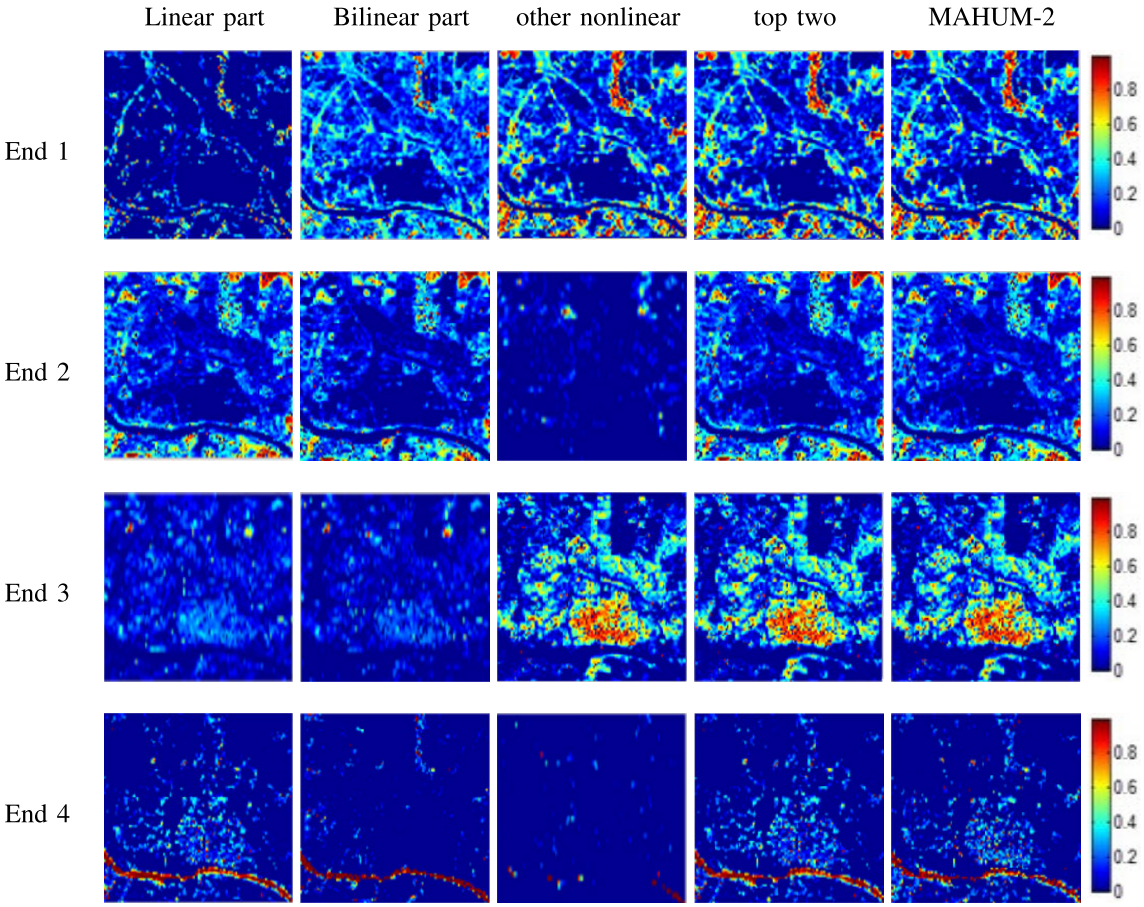


Fig. 16. Results of the analysis of the Pavia PRISMA dataset by means of the MAHUM-2 algorithm. The linear, bilinear, and other nonlinear parts are the quantitative analysis results of the three types of unmixing methods proposed in this article. (Top two) Result of adding the two items with the most unmixing components according to their weights. MAHUM refers to the result of all unmixing components.

TABLE V
PAVIA DATASET EVALUATION INDEXES

	N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU	3D-CNN	MAHUM-1	MAHUM-2
RE	0.2059	0.1858	0.1555	0.1733	0.1693	0.1351	0.1428	0.0826	0.0719	0.0725	**0.0635**

TABLE VI
COMPUTATIONAL TIMES FOR DIFFERENT DATASETS

	Synthetic	Moffett Field	Jasper Ridge	Samson Ridge	Pavia
Full data(s)	866	243	298	221	375
Train(s)	316	93	127	114	169
Test(s)	59	19	28	25	38

DMBU's tree and water performance, compared to MAHUM-1's. The visual results also meet expectations.

Fig. 14 shows several classical methods' comparison and display results. From the visual effect, Many classic methods do not perform well in the river section. In addition, buildings and trees often exhibit misclassification. In these difficulties, MAHUM-1 performs best on water. MAHUM-2 performs best in architecture. This fully demonstrates the effectiveness of our method for complex urban environments.

Figs. 15 and 16 are diagrams showing the quantitative results of MAHUM-1 and MAHUM-2. From Figs. 15 and 16, different components have shown emphasis in terms of methods. We can intuitively observe that water is more suitable for linear mixing methods, while buildings and soil are more suitable for nonlinear methods, which is consistent with expert experience.

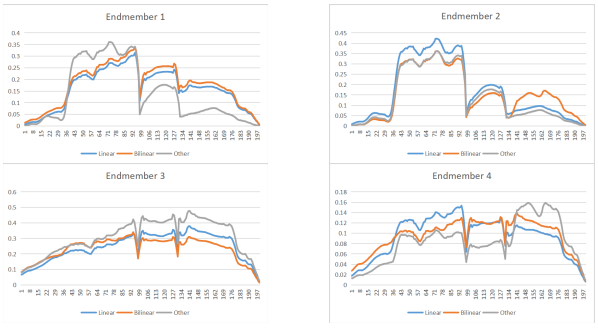


Fig. 17. Estimated nominal endmember signatures extracted from the Pavia PRISMA dataset.

From Fig. 17, we can see that although we decompose the mixed pixels through three methods. However, the spectral loss of the endmembers is not severe. The vast majority of the spectra remain within a controllable range. A small amount

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of the spectrum is disturbed due to the influence of complex environments.

5) Execution Time: In this algorithm test process, the results were performed based on a computer that has Intel[1] Core[2] $i7-8700K$ CPU (3.70 GHZ), 32 GB RAM, and 1080 Ti GPU. Table VI shows the running times under the same experiment conditions.

From Table VI, it can be seen that more complex data requires more training and final time. However, our algorithm is basically controlled from 200 to 400 s in the test dataset, which is acceptable in practical applications.

V. Conclusion

In this article, to solve the problem that the current mainstream work ignores the multiple reflections of a single material, MAHUM, which combines various models through a multitask framework, is proposed. This method considers several mainstream anti-mixing models at present, and each method can play its characteristics through the multitask mode to improve the experiment accuracy. In addition, this algorithm can quantitatively analyze the adaptability of the same substance in different unmixing methods from the results. This article visually demonstrates each method's ability to unmix other materials in experiments. The results reached the expected level.

Of course, the current method still needs to be improved. First, this method does not reasonably explain other nonlinear parts but uses the deep learning black box mode. Second, the content of this part starts from a different perspective than higher-order nonlinear unmixing, which makes the research on multiple scattering of the algorithm still insufficient. Third, the weight calculation of each algorithm needs to be simplified, which also affects the accuracy.

Acknowledgment

The used PRISMA Product by the Italian Space Agency (ASI), was delivered under a license to use.

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